

# Decision-Making Under Uncertainty: Exam Solutions

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## Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

## Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

## Evaluated criteria

- C1** Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.
- C2** Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.
- C3** Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.
- C4** Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

## 1 Community Festival Beverage Booth

[C1-1    C2-2]

A community festival company must decide which beverage booth to operate before knowing how strong customer attendance will be. The company can choose either a basic booth or a premium booth, while festival attendance may turn out to be high or low. If the company chooses the basic booth, it expects to earn a profit of sixteen thousand US dollars when attendance is high and ten thousand US dollars when attendance is low. If it chooses the premium booth, it expects to earn a profit of twenty-nine thousand US dollars when attendance is high and five thousand US dollars when attendance is low.

### Questions/Tasks.

**C1** Characterize the key elements of the decision-making situation.

[1]

**C2** Construct the payoff table.

[2]

## C1 - Characterization

[1]

| Element        | Description                                                   |
|----------------|---------------------------------------------------------------|
| Alternatives   | <b>A1:</b> Basic booth <b>A2:</b> Premium booth               |
| Statesofnature | <b>S1:</b> High attendance <b>S2:</b> Low attendance          |
| Events         | Actual festival attendance after the booth decision           |
| Consequences   | Profit in thousand US dollars for each alternative–state pair |

## C2 - Payoff table

[2]

Each stated consequence is placed at the intersection of its alternative and state:

$$\begin{aligned} P(A_1, S_1) &= 16, & P(A_1, S_2) &= 10, \\ P(A_2, S_1) &= 29, & P(A_2, S_2) &= 5. \end{aligned}$$

Thus, in thousand US dollars, the completed payoff table is

| Alternative         | High $S_1$ | Low $S_2$ |
|---------------------|------------|-----------|
| Basic booth $A_1$   | 16         | 10        |
| Premium booth $A_2$ | 29         | 5         |



## 2 Local Delivery Fleet

[C1-1    C2-3]

A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic.



**Final payoff table (profit in thousand CNY).**

| Fleet              | Clear traffic | Congested traffic |
|--------------------|---------------|-------------------|
| Bicycle fleet      | 61            | 36                |
| Electric-van fleet | 77            | 44                |



### 3 Regional Food Delivery Service

[C1-2 C2-5]

A regional food delivery company must choose between a standard delivery service and an express delivery service before customer demand is known. Demand may be high or low. The standard service earns 3 thousand GBP per completed delivery order and also provides a fixed operating contribution of 5 thousand GBP. The express service earns 5 thousand GBP per completed delivery order and provides a fixed operating contribution of 8 thousand GBP. High demand adds another 2 thousand GBP per completed order to either service, while low demand adds 1 thousand GBP per completed order. The standard service is expected to complete 15 orders under high demand and 9 orders under low demand; the express service is expected to complete 11 orders under high demand and 7 orders under low demand.

**Questions/Tasks.**

**C1** Characterize the key elements of the decision-making situation.

[2]

**C2** Construct the payoff table.

[5]

#### C1 - Characterization

[2]

| Element          | Description                                                              |
|------------------|--------------------------------------------------------------------------|
| Alternatives     | <b>A1:</b> Standard delivery service <b>A2:</b> Express delivery service |
| States of nature | <b>S1:</b> High customer demand <b>S2:</b> Low customer demand           |
| Events           | Actual customer demand after the delivery service is selected            |
| Consequences     | Profit in thousand GBP for each alternative–state pair                   |

#### C2 - Payoff table

[5]

**Relevant values.**

| Abstract component                    | Context component                                                           | Math symbol   | Corresponding value                                                                                       |
|---------------------------------------|-----------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------|
| Quantity, alternative–state dependent | Expected completed delivery orders for each service under each demand level | $q(A_i, S_j)$ | Standard–high: 15 orders;<br>Standard–low: 9 orders;<br>Express–high: 11 orders;<br>Express–low: 7 orders |
| Variable, alternative dependent       | Contribution per completed delivery order from the chosen service           | $v^A(A_i)$    | Standard service: 3 thousand GBP/order; Express service: 5 thousand GBP/order                             |
| Variable, state dependent             | Additional contribution per completed order from customer demand            | $v^S(S_j)$    | High demand: 2 thousand GBP/order; Low demand: 1 thousand GBP/order                                       |
| Fixed, alternative dependent          | Fixed operating contribution from the chosen service                        | $f^A(A_i)$    | Standard service: 5 thousand GBP; Express service: 8 thousand GBP                                         |

**Payoff equation.**

$$P_{ay}(A_i, S_j) = q(A_i, S_j) [v^A(A_i) + v^S(S_j)] + f^A(A_i).$$

**Calculations (thousand GBP).**

| Service          | High customer demand | Low customer demand |
|------------------|----------------------|---------------------|
| Standard service | $15(3 + 2) + 5 = 80$ | $9(3 + 1) + 5 = 41$ |
| Express service  | $11(5 + 2) + 8 = 85$ | $7(5 + 1) + 8 = 50$ |

**Final payoff table (profit in thousand GBP).**

| Service          | High customer demand | Low customer demand |
|------------------|----------------------|---------------------|
| Standard service | 80                   | 41                  |
| Express service  | 85                   | 50                  |



## 4 Local Caf'e Investment

[C1-1 C3-2]

A small caf'e must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in million COP. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.

**Questions/Tasks.**

**C1** Characterize the key elements of the decision-making situation.

[1]

**C3** Use revenue and costs to construct the payoff table.

[2]

## C1 - Characterization

[1]

| Element          | Description                                                 |
|------------------|-------------------------------------------------------------|
| Alternatives     | <b>A1:</b> Launch dessert line <b>A2:</b> Keep current menu |
| States of nature | <b>S1:</b> High traffic <b>S2:</b> Low traffic              |
| Events           | Actual foot traffic after the menu decision                 |
| Consequences     | Profit in million COP for each alternative–state pair       |

## C3 - Calculating profit and building the payoff table

[2]

Relevant values.

| Abstract component                   | Context component                                                          | Math symbol   | Corresponding value                                                                                                                     |
|--------------------------------------|----------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Revenue, alternative–state dependent | Total revenue generated by each menu decision under each traffic condition | $R(A_i, S_j)$ | Dessert–high: 58 million COP;<br>Dessert–low: 27 million COP;<br>Current menu–high: 49 million COP;<br>Current menu–low: 41 million COP |
| Cost, alternative–state dependent    | Total cost associated with each menu decision under each traffic condition | $C(A_i, S_j)$ | Dessert–high: 16 million COP;<br>Dessert–low: 19 million COP;<br>Current menu–high: 21 million COP;<br>Current menu–low: 21 million COP |

Profit equation.

$$P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Calculations (million COP).

| Alternative and state     | Revenue | Cost | Profit         |
|---------------------------|---------|------|----------------|
| Dessert line–high traffic | 58      | 16   | $58 - 16 = 42$ |
| Dessert line–low traffic  | 27      | 19   | $27 - 19 = 8$  |
| Current menu–high traffic | 49      | 21   | $49 - 21 = 28$ |
| Current menu–low traffic  | 41      | 21   | $41 - 21 = 20$ |

Final payoff table (profit in million COP).

| Alternative         | High traffic | Low traffic |
|---------------------|--------------|-------------|
| Launch dessert line | 42           | 8           |
| Keep current menu   | 28           | 20          |



## 5 Custom Furniture Product Line

[C1-1 C3-3]

A custom-furniture workshop must choose between producing dining tables and producing office desks before market conditions are known. Conditions may be strong demand with favourable material prices or weak demand with higher material prices. The respective quantities are 7 and 4 production lots for dining tables, and 6 and 3 lots for office desks. Revenue per lot depends on the product: 14 thousand EUR for dining tables and 16 thousand EUR for office desks. Cost per lot depends on the state: 6 thousand EUR under strong demand and favourable prices, and 9 thousand EUR under weak demand and higher prices.

### Questions/Tasks.

**C1** Characterize the key elements of the decision-making situation.

[1]

**C3** Use revenue and costs to construct the payoff table.

[3]

### C1 - Characterization

[1]

| Element          | Description                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------|
| Alternatives     | <b>A1:</b> Produce dining tables <b>A2:</b> Produce office desks                                          |
| States of nature | <b>S1:</b> Strong demand and favourable material prices <b>S2:</b> Weak demand and higher material prices |
| Events           | Market conditions revealed after the production decision                                                  |
| Consequences     | Profit in thousand EUR for each product–market–state pair                                                 |

### C3 - Profit and payoff table

[3]



### Relevant values.

| Abstract component                    | Context component                                            | Math symbol   | Corresponding value                                                   |
|---------------------------------------|--------------------------------------------------------------|---------------|-----------------------------------------------------------------------|
| Quantity, alternative–state dependent | Production lots for each product under each market condition | $q(A_i, S_j)$ | Tables–strong: 7; Tables–weak: 4; Desks–strong: 6; Desks–weak: 3      |
| Variable, alternative dependent       | Revenue per production lot                                   | $r(A_i)$      | Dining tables: 14 thousand EUR/lot; Office desks: 16 thousand EUR/lot |
| Variable, state dependent             | Cost per production lot                                      | $c(S_j)$      | Strong demand: 6 thousand EUR/lot; Weak demand: 9 thousand EUR/lot    |

### Profit equation.

$$R = q(A_i, S_j)r(A_i), \quad C = q(A_i, S_j)c(S_j), \quad P_{ay}(A_i, S_j) = R - C.$$

### Calculations (thousand EUR).

| Cell         | Revenue      | Cost        | Profit         |
|--------------|--------------|-------------|----------------|
| $(A_1, S_1)$ | $7(14) = 98$ | $7(6) = 42$ | $98 - 42 = 56$ |
| $(A_1, S_2)$ | $4(14) = 56$ | $4(9) = 36$ | $56 - 36 = 20$ |
| $(A_2, S_1)$ | $6(16) = 96$ | $6(6) = 36$ | $96 - 36 = 60$ |
| $(A_2, S_2)$ | $3(16) = 48$ | $3(9) = 27$ | $48 - 27 = 21$ |

**Final payoff table (profit in thousand EUR).**

| Alternative           | Strong demand, favourable prices | Weak demand, higher prices |
|-----------------------|----------------------------------|----------------------------|
| Produce dining tables | 56                               | 20                         |
| Produce office desks  | 60                               | 21                         |

## 6 Farm Subscription Plans

[C1-2 C3-5]

A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 8 and 5 for the standard box and 6 and 4 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 9 and 12 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 7 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 10 thousand USD under strong demand and 14 thousand USD under weak demand.

**Questions/Tasks.**

**C1** Characterize the key elements of the decision-making situation.

[2]

**C3** Use revenue and costs to construct the payoff table.

[5]

### C1 - Characterization

[2]

| Element          | Description                                                                    |
|------------------|--------------------------------------------------------------------------------|
| Alternatives     | <b>A1:</b> Offer a standard produce box <b>A2:</b> Offer a premium produce box |
| States of nature | <b>S1:</b> Strong seasonal demand <b>S2:</b> Weak seasonal demand              |
| Events           | Seasonal market conditions revealed after the subscription-plan decision       |
| Consequences     | Profit in thousand USD for each plan–market-state pair                         |

### C3 - Profit and payoff table

[5]



**Relevant values.**



| Abstract component                    | Context component                                                      | Math symbol   | Corresponding value                                                                                         |
|---------------------------------------|------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------|
| Quantity, alternative–state dependent | Subscription lots for each plan under each demand level                | $q(A_i, S_j)$ | Standard–strong: 8;<br>Standard–weak: 5;<br>Premium–strong: 6;<br>Premium–weak: 4 hundreds of subscriptions |
| Variable, alternative dependent       | Plan-based revenue rate per hundred subscriptions                      | $r^A(A_i)$    | Standard box: 9 thousand USD/100 subscriptions; Premium box: 12 thousand USD/100 subscriptions              |
| Variable, state dependent             | Additional revenue rate per hundred subscriptions from seasonal demand | $r^S(S_j)$    | Strong demand: 3 thousand USD/100 subscriptions; Weak demand: 1 thousand USD/100 subscriptions              |
| Variable, alternative dependent       | Plan-based cost rate per hundred subscriptions                         | $c^A(A_i)$    | Standard box: 4 thousand USD/100 subscriptions; Premium box: 7 thousand USD/100 subscriptions               |
| Fixed, state dependent                | Fixed distribution cost under each demand condition                    | $f_c(S_j)$    | Strong demand: 10 thousand USD; Weak demand: 14 thousand USD                                                |

**Profit equation.**

$$R = q(A_i, S_j) [r^A(A_i) + r^S(S_j)], \quad C = q(A_i, S_j)c^A(A_i) + f_c(S_j), \quad P_{ay}(A_i, S_j) = R - C.$$

**Calculations (thousand USD).**

| Cell         | Revenue          | Cost             | Profit         |
|--------------|------------------|------------------|----------------|
| $(A_1, S_1)$ | $8(9 + 3) = 96$  | $8(4) + 10 = 42$ | $96 - 42 = 54$ |
| $(A_1, S_2)$ | $5(9 + 1) = 50$  | $5(4) + 14 = 34$ | $50 - 34 = 16$ |
| $(A_2, S_1)$ | $6(12 + 3) = 90$ | $6(7) + 10 = 52$ | $90 - 52 = 38$ |
| $(A_2, S_2)$ | $4(12 + 1) = 52$ | $4(7) + 14 = 42$ | $52 - 42 = 10$ |

**Final payoff table (profit in thousand USD).**

| Alternative          | Strong seasonal demand | Weak seasonal demand |
|----------------------|------------------------|----------------------|
| Standard produce box | 54                     | 16                   |
| Premium produce box  | 38                     | 10                   |

## 7 New Device Launch Strategy

[C4-2]

A technology company must choose a strategy for launching a new wearable device. Market demand could be high, medium, or low. All payoffs are in thousands of dollars.

| Alternative     | Demand |        |     |
|-----------------|--------|--------|-----|
|                 | High   | Medium | Low |
| Premium launch  | 90     | 38     | -25 |
| Standard launch | 65     | 45     | 20  |
| Economy launch  | 42     | 34     | 27  |
| Delay launch    | 18     | 18     | 18  |

### Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? [2]

### C4 - Maximax and Maximin

[2]



Complete the table using letters first.

Replace the letters with their values.

|     | $A_1$ | $A_2$ | $A_3$ | $A_4$ | Decision                     |
|-----|-------|-------|-------|-------|------------------------------|
| Max | $M_1$ | $M_2$ | $M_3$ | $M_4$ | $A_{\text{Maximax}}^* = A_1$ |
| Min | $m_1$ | $m_2$ | $m_3$ | $m_4$ | $A_{\text{Maximin}}^* = A_3$ |

|     | $A_1$ | $A_2$ | $A_3$ | $A_4$ | Decision                     |
|-----|-------|-------|-------|-------|------------------------------|
| Max | 90    | 65    | 42    | 18    | $A_{\text{Maximax}}^* = A_1$ |
| Min | -25   | 20    | 27    | 18    | $A_{\text{Maximin}}^* = A_3$ |

#### Maximax

Therefore, the optimistic decision is

$$M_1 = \max\{90, 38, -25\} = 90$$

$$A_{\text{Maximax}}^* = A_1 \quad (\text{Premium launch})$$

$$M_2 = \max\{65, 45, 20\} = 65$$

and the conservative decision is

$$M_3 = \max\{42, 34, 27\} = 42$$

$$M_4 = \max\{18, 18, 18\} = 18$$

$$A_{\text{Maximin}}^* = A_3 \quad (\text{Economy launch}).$$

#### Maximin

$$m_1 = \min\{90, 38, -25\} = -25$$

$$m_2 = \min\{65, 45, 20\} = 20$$

$$m_3 = \min\{42, 34, 27\} = 27$$

$$m_4 = \min\{18, 18, 18\} = 18$$

## 8 Restaurant Expansion Strategy

[C4-2]

A restaurant chain must choose a strategy for expanding its operations. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

| Alternative        | Customer demand |        |     |
|--------------------|-----------------|--------|-----|
|                    | High            | Medium | Low |
| Large expansion    | 85              | 32     | -18 |
| Moderate expansion | 62              | 44     | 16  |
| Small expansion    | 45              | 36     | 24  |
| No expansion       | 20              | 20     | 20  |

### Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? [2]

### C4 - Maximax and Maximin

[2]



Complete the table using letters first.

Replace the letters with their values.

|     | $A_1$ | $A_2$ | $A_3$ | $A_4$ | Decision                     |
|-----|-------|-------|-------|-------|------------------------------|
| Max | $M_1$ | $M_2$ | $M_3$ | $M_4$ | $A_{\text{Maximax}}^* = A_1$ |
| Min | $m_1$ | $m_2$ | $m_3$ | $m_4$ | $A_{\text{Maximin}}^* = A_3$ |

|     | $A_1$ | $A_2$ | $A_3$ | $A_4$ | Decision                     |
|-----|-------|-------|-------|-------|------------------------------|
| Max | 85    | 62    | 45    | 20    | $A_{\text{Maximax}}^* = A_1$ |
| Min | -18   | 16    | 24    | 20    | $A_{\text{Maximin}}^* = A_3$ |

#### Maximax

Therefore, the optimistic decision is

$$M_1 = \max\{85, 32, -18\} = 85$$

$$A_{\text{Maximax}}^* = A_1 \quad (\text{Large expansion})$$

$$M_2 = \max\{62, 44, 16\} = 62$$

and the conservative decision is

$$M_3 = \max\{45, 36, 24\} = 45$$

$$M_4 = \max\{20, 20, 20\} = 20$$

$$A_{\text{Maximin}}^* = A_3 \quad (\text{Small expansion}).$$

#### Maximin

$$m_1 = \min\{85, 32, -18\} = -18$$

$$m_2 = \min\{62, 44, 16\} = 16$$

$$m_3 = \min\{45, 36, 24\} = 24$$

$$m_4 = \min\{20, 20, 20\} = 20$$

## 9 Renewable Energy Investment Strategy

[C4-2]

A renewable-energy company must choose a strategy for developing a new solar project. Market conditions could be very favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

| Alternative         | Market conditions |            |        |              |
|---------------------|-------------------|------------|--------|--------------|
|                     | Very favourable   | Favourable | Stable | Unfavourable |
| Large solar farm    | 110               | 55         | 10     | -35          |
| Medium solar farm   | 82                | 60         | 28     | 5            |
| Small solar farm    | 58                | 48         | 35     | 18           |
| Partnership project | 45                | 40         | 32     | 25           |

### Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? [2]

### C4 - Maximax and Maximin

[2]



Complete the table using letters first.

Replace the letters with their values.

|     | $A_1$ | $A_2$ | $A_3$ | $A_4$ | Decision                     |
|-----|-------|-------|-------|-------|------------------------------|
| Max | $M_1$ | $M_2$ | $M_3$ | $M_4$ | $A_{\text{Maximax}}^* = A_1$ |
| Min | $m_1$ | $m_2$ | $m_3$ | $m_4$ | $A_{\text{Maximin}}^* = A_4$ |

|     | $A_1$ | $A_2$ | $A_3$ | $A_4$ | Decision                     |
|-----|-------|-------|-------|-------|------------------------------|
| Max | 110   | 82    | 58    | 45    | $A_{\text{Maximax}}^* = A_1$ |
| Min | -35   | 5     | 18    | 25    | $A_{\text{Maximin}}^* = A_4$ |

#### Maximax

Therefore, the optimistic decision is

$$M_1 = \max\{110, 55, 10, -35\} = 110$$

$$A_{\text{Maximax}}^* = A_1 \quad (\text{Large solar farm})$$

$$M_2 = \max\{82, 60, 28, 5\} = 82$$

and the conservative decision is

$$M_3 = \max\{58, 48, 35, 18\} = 58$$

$$M_4 = \max\{45, 40, 32, 25\} = 45$$

$$A_{\text{Maximin}}^* = A_4 \quad (\text{Partnership project}).$$

#### Maximin

$$m_1 = \min\{110, 55, 10, -35\} = -35$$

$$m_2 = \min\{82, 60, 28, 5\} = 5$$

$$m_3 = \min\{58, 48, 35, 18\} = 18$$

$$m_4 = \min\{45, 40, 32, 25\} = 25$$